Practical work #4 (6 hours)
Development of a social network database

1. ***Development of a social network database.***

Using the Neo4j graph DBMS, develop a database for a model social network. A social network node has the following attributes: full name, gender, age, city, list of interest groups (for example: sports, games, computers). The semantics of the connection between nodes is the friendly relations of the corresponding persons.

Instructions for the task 1.

Instructions for Task 1:
Develop a script with Cypher queries to load the social network database into the Neo4j DBMS. The database must contain at least 10 nodes and at least 15 connections between them.

Load the database into the Neo4j DBMS using the console client (Neo4jShell.bat).

2. ***Graph query development.***

Develop and test data sampling queries from the created graph database.

Instructions for the task 2.

Developing Graph Queries:

2.1. Study reference information on the Cypher language regarding Reading Clauses and develop the following queries:

1. Display a list of names and ages of men, sorted in descending order by age.
2. Display an ordered list of the names of a given person's friends.
3. Display an ordered list of the names of friends of the given person's friends.
4. Display an alphabetically ordered list of persons' names along with the number of friends each person has.

2.2. Study the background information on the Cypher language regarding Functions and develop the following queries:

1. Display an ordered list of social network groups.
2. Display an ordered list of groups of the person with the given full name.
3. Display a list of social network groups, showing the number of members in each group, ordered in descending order of membership count.
4. Display a list of full names of persons with the number of groups they belong to, ordered in descending order by the number of groups.
5. Display the total number of groups that include friends of friends of the person with the given full name.

3. ***Graph visualization.***

Develop and test queries to update the properties of graph nodes and use them to visualize the graph.

Instructions for the task 3.

1. Study the reference information on Neo4j visualization tools. Display the graph.

2. Add a *friend* property to all friends of specific person and configure the visualization profile so that nodes with the *friend* property are displayed in red.

3. Add the *twoHandFriend* property to all friends of friends of the same person and configure the visualization profile so that nodes with the *twoHandFriend* property are displayed in yellow.

4. Add the *manyFriends* property to persons with more than 5 friends, the *fewFriends* property to persons with less than 3 friends. Configure the visualization profile so that nodes with the *manyFriends* property are displayed in green, nodes with the *fewFriends* property are displayed in red, and all other nodes are displayed in yellow.

5. Add the *group1* property to persons belonging to some group, *group2* to persons belonging to another group, and the *bothGroups* property to persons belonging to both groups. Configure the visualization profile so that nodes with the *group1* property are displayed in blue, nodes with the *group2* property are displayed in red, and nodes with the *bothGroups* property are displayed in purple.

Protection of practical work

The defense of the practical work occurs only after its completion. During the defense, the student:

1) explains the progress of the task;

3) explains the results obtained as a result of completing the task.